

Supplementary Information: Unveiling Mycobacterium marinum Cellular and Metabolic Remodeling Under Simulated Microgravity

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1 Supplementary Tables Overview

All empirical data tables are archived in the repository under **results_tables/** and available via Zenodo:

- **Table S1** (`Table_S1_raw_counts_matrix.tsv`): Complete raw transcript count matrix (5,510 genes $\times$ 9 samples) generated by kallisto v0.52.0 pseudoalignment of NextSeq 550 paired-end reads.
- **Table S2** (`Table_S2_normalized_log2cpm_expression.tsv`): *Normalized* $\log_2(\text{CPM}+1)$ matrix across 4,964 expressed genes.
- **Table S3** (`Table_S3_sample_metadata.tsv`): Experimental metadata for NASA OSDR OSD-528.
- **Table S4** (`Table_S4_differential_expression_3dclinostat_vs_1g.tsv`): 351 significant DEGs in 3D Clinostat vs 1g.
- **Table S5** (`Table_S5_differential_expression_rpm2_vs_1g.tsv`): 738 significant DEGs in RPM2.0 vs Static 1g.
- **Table S6** (`Table_S6_differential_expression_3dclinostat_vs_rpm2.tsv`): 242 DEGs comparing simulation kin.
- **Table S7** (`Table_S7_wgcna_module_assignments_goslim.tsv`): WGCNA module assignments, standardized hub scores.
- **Table S8** (`Table_S8_wgcna_module_eigenvalues.tsv`): Module Eigenvalue profiles across all 9 samples.
- **Table S9** (`Table_S9_wgcna_module_trait_correlations.tsv`): Pearson correlation coefficients and asymptotic p-values.

- Table S10 (Table_S10_goslim_pathway_enrichment.tsv) : GOSlim pathway over-representation metrics (k /r, Hochberg FDR).
- Table S11 (Table_S11_full_gene_ontology_enrichment.tsv) : Comprehensive GO enrichment table across Biological Processes.
- Table S12 (Table_S12_tabPFN_loocv_predictions.tsv) : TabPFN leave-one-out cross-validation predictions.
- Table S13 (Table_S13_tabPFN_feature_importance.tsv) : TabPFN permutation feature importance rankings.
- Table S14 (Table_S14_cellular_metabolic_model_reactions.tsv) : Reaction-level GPR mapping across 32 enzymatic reactions.
- Table S15 (Table_S15_metabolic_subsystem_perturbation_index.tsv) : Quantitative Subsystem Perturbation Index (SPI) ranking.
- Table S16 (Table_S16_pan_microbial_osdr_compndium_78_studies.tsv) : Compendium of 78 microbial space flight studies.
- Table S17 (Table_S17_cellular_metabolic_model_sbml.json) : SBML-compatible computational model schema.

2 Supplementary Figures Overview

All supplementary figures are archived as vector PDFs and high-resolution 300 DPI PNGs in [supplementary_figures/](#):

- **Figure S1:** Systems architecture, biological model, simulation hardware, and analytical framework.
- **Figure S2:** Transcriptomic PCA biplot (70.1% PC1 variance) and volcano plot of 351 DEGs.
- **Figure S3:** Gene count distribution across 5 GOSlim co-expression modules and module-trait heatmap.
- **Figure S4:** TabPFN tabular foundation model benchmark under LOOCV, confusion matrix, and feature importances.
- **Figure S5:** Enriched GOSlim functional pathways with calibrated FAIR Blue-White-Red color fill.
- **Figure S6:** Intramodular hub connectivity (k_{within} vs k_{total}) and inter-module regulatory interactome.
- **Figure S7:** Pan-microbial spaceflight meta-analysis landscape across 78 OSDR studies and concordance matrix.
- **Figure S8:** Hexagonal trajectory radar mapping phenotypic concordance across 6 physiological axes.
- **Figure S9:** Multi-compartment cellular cross-section and Subsystem Perturbation Index (SPI) ranking.

3 Read Quality Control Pseudoalignment Convergence

Pseudoalignment against *M. marinum* M strain (NC_010612.1) converged with high pseudoalignment efficiency across all 9 biological replicates (mean 1.21 million reads mapped per sample). Normalized counts were converted to log-CPM values using edgeR TMM library scaling. Zero synthetic data were utilized in any calculation.